Karim El Yaagoubi

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PROFILE

Software & Data Engineer passionate about building scalable applications, backend systems, and low-level software.

PROFESSIONAL EXPERIENCE

Astound Technology, *Backend Software Engineer intern* Jul 2026 – Oct 2026 | Casablanca

- Developed RESTful API using **Spring Boot** following layered architecture and clean coding practices.
- Designed and implemented data models with **MongoDB** and **PostgreSQL**.
- Integrated authentication and authorization using **Keycloak** with JWT and role based access control.
- Integrated API with **Next.js** applications.

PROJECTS

Goldfinch Sep 2026 – Present

Goldfinch is a small RDF triplestore written in C. It stores RDF triples in memory and supports basic SPARQL queries with prefixes, variables, IRIs, and triple-pattern matching.

Semantic Modeling & Analytics of Cancer Data | Python, RDF, SPARQL, OWL, *Academic project* Feb 2026 – Jun 2026

- Designed a custom OWL ontology modeling breast cancer knowledge across 5 classes (Patient, Diagnosis, Treatment, Gene Expression, Mutation)
- Built a Python semantic lifting pipeline transforming the METABRIC dataset (1,904 patients × 693 variables) into an RDF knowledge graph of ~3.85M triples
- Benchmarked AllegroGraph, Blazegraph, and Virtuoso in isolated Docker containers on query latency, RAM, and disk.

Audio Tag Editor | Dart, Java, Flutter, BLoC, Android Aug 2025

- Developed and maintained an **open-source Flutter plugin** for reading and editing audio metadata
- Implemented **platform-specific Android code (Java)** and bridged it using method channels
- Supported multiple formats: MP3, M4A, FLAC, OGG, AIFF, WAV, WMA

- Built and launched a **production-ready Flutter mobile application** for audio metadata editing
- Integrated a custom Flutter plugin with native Android (Java) code for efficient file processing
- Implemented **BLoC architecture** for scalable and maintainable state management
- Achieved **10,000+ installs** and **60+ paying users**, generating real revenue on Google Play

CourierFlow | Flutter, Spring Boot, GraphHopper, JWT May 2025

- Developed a **Flutter mobile application** for delivery management and route tracking
- Integrated with a **Spring Boot backend API** implementing GraphHopper for route optimization
- Implemented **JWT-based authentication** for secure communication between client and backend
- Enabled efficient route planning and delivery workflow management

CERTIFICATES

- Oracle Certified Professional: Java SE 17 Developer

SKILLS

Languages — Java, Dart, C, PL/SQL, **Backend** — Spring Boot, REST APIs, JWT, **Data Engineering & Semantic Web** — SPARQL, RDF, OWL, Knowledge Graphs, AllegroGraph, Blazegraph, Virtuoso, **Cloud & DevOps** — Docker, Linux, Git, GitHub, **Mobile** — Flutter, **Core** — Data Structures, Algorithms, Object-Oriented Design

EDUCATION

National School Of Applied Sciences, *Data science and cloud computing*

Les Classes Préparatoires aux Grandes Écoles (CPGE), *Technology and industrial science*